

Artificial Intelligence and the Skill Premium

An Endogenous Growth Perspective

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Introduction

- ▶ AI transforms production and labor markets
- ▶ Labor share in national income and the employment to population ratio are declining (U.S. Bureau of Labor Statistics)
- ▶ Similar claims have been made, but have not always come true
 - ▶ New technologies would create widespread technological unemployment as machines replaced human labor (Keynes, 1930)
 - ▶ As machines continue to invade society, human labor is gradually rendered redundant (Robert Heilbroner, 1965)
 - ▶ Leontief's analogy: the technologies of the early twentieth century made horses redundant

Mechanisms of AI

- ▶ Substitution
 - ▶ Automation in terms of industrial robots substitutes routine mechanical tasks
 - ▶ AI predominantly substitutes for tasks performed by high-skill workers
- ▶ Task creation
 - ▶ The Second Industrial Revolution generated jobs for engineers, machinists, repairmen, conductors, back-office workers, and managers
 - ▶ AI has also generated new jobs like audio-visual specialists and data administrators
- ▶ Learning

Model Overview

- ▶ Task space: $i \in [N(t) - 1, N(t)]$
- ▶ Final output:

$$Y(t) = \left(\int_{N(t)-1}^{N(t)} y(i, t)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}}$$

- ▶ Technology margins:
 - ▶ $I(t)$: automation
 - ▶ $N(t)$: task creation

AI Technology Input

- ▶ Each task uses AI input $q(i, t)$
- ▶ Interpretation:
 - ▶ Algorithms / models / software
- ▶ Market structure:
 - ▶ Monopoly producer, marginal cost $\mu\psi$ ($\mu < 1$)
 - ▶ Competitive fringe cost ψ
- ▶ Equilibrium price:

$$p_q(i, t) = \psi$$

- ▶ AI is factor-augmenting

Intermediate Production

$$y(i, t) = \begin{cases} q(i, t)^\eta [k(i, t) + \gamma_L(i, t)l(i, t) + \gamma_H(i, t)h(i, t)]^{1-\eta}, & i \leq l(t) \\ q(i, t)^\eta [\gamma_L(i, t)l(i, t) + \gamma_H(i, t)h(i, t)]^{1-\eta}, & i > l(t) \end{cases}$$

- Capital substitutes labor only for $i \leq l(t)$

Unit Cost Function

$$\min_{q,x} \quad \psi q + x \quad \text{s.t.} \quad y = q^\eta x^{1-\eta}$$

$$c(x) = \eta^{-\eta} (1 - \eta)^{\eta-1} \psi^\eta x^{1-\eta}$$

$$x \in \left\{ R, \frac{W_L}{\gamma_L}, \frac{W_H}{\gamma_H} \right\}$$

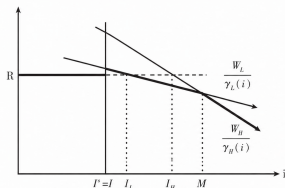
Price Function

If $i \leq I(t)$:

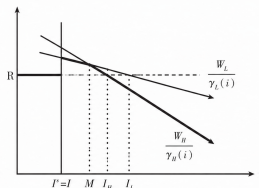
$$p(i, t) = \min \left\{ c(R), c \left(\frac{W_L}{\gamma_L(i, t)} \right), c \left(\frac{W_H}{\gamma_H(i, t)} \right) \right\}$$

If $i > I(t)$:

$$p(i, t) = \min \left\{ c \left(\frac{W_L}{\gamma_L(i, t)} \right), c \left(\frac{W_H}{\gamma_H(i, t)} \right) \right\}$$



(a) Low-skill labor as the main determinant of investment



(b) High-skill labor as the main determinant of investment

Market Clearing

$$(1 - \eta)y(1 - n)c(R)^{1-\sigma}R^{-1} = K$$

$$(1 - \eta)y \int_0^m c\left(\frac{w_L}{\gamma_L}\right)^{1-\sigma} w_L^{-1} di = L$$

$$(1 - \eta)y \int_m^n c\left(\frac{w_H}{\gamma_H}\right)^{1-\sigma} w_H^{-1} di = H$$

Assumption 1: Task-Dependent Productivity

Labor productivity functions satisfy:

$$\frac{\partial \gamma_L(i, t)}{\partial i} > 0, \quad \frac{\partial \gamma_H(i, t)}{\partial i} > 0$$

$$\frac{\partial}{\partial i} \left(\frac{\gamma_H(i, t)}{\gamma_L(i, t)} \right) > 0$$

Implications:

- ▶ Productivity increases with task complexity i
- ▶ Labor dominates capital in high-complexity tasks
- ▶ High-skill labor has comparative advantage in complex tasks

Factor Prices and Thresholds

Effective factor costs:

$$\text{Capital: } R(t), \quad \text{Low-skill: } \frac{W_L(t)}{\gamma_L(i, t)}, \quad \text{High-skill: } \frac{W_H(t)}{\gamma_H(i, t)}$$

Automation thresholds:

$$\frac{W_L(t)}{\gamma_L(I_L(t), t)} = R(t), \quad \frac{W_H(t)}{\gamma_H(I_H(t), t)} = R(t)$$

Binding automation frontier:

$$\tilde{I}(t) = \min\{I(t), I_L(t), I_H(t)\}$$

Interpretation:

- ▶ Capital replaces labor only when cost-efficient
- ▶ Automation is jointly determined by technology and wages

Skill Allocation and Task Sorting

Indifference condition:

$$\frac{W_L(t)}{\gamma_L(M(t), t)} = \frac{W_H(t)}{\gamma_H(M(t), t)}$$

Task allocation:

- ▶ $i < M(t)$: low-skill labor minimizes cost
- ▶ $i > M(t)$: high-skill labor minimizes cost

Economic interpretation:

- ▶ Sorting driven by comparative advantage
- ▶ Higher $M(t) \Rightarrow$ expansion of low-skill task range

Assumption 2: Interior Automation

- ▶ Restriction:

$$I^*(t) = \min\{I(t), I_L(t), I_H(t), M(t)\} = I(t)$$

- ▶ Ensures:
 - ▶ Automation is active but not dominant
 - ▶ Labor remains essential in production
- ▶ Avoids corner solutions:
 - ▶ Full automation (degenerate equilibrium)

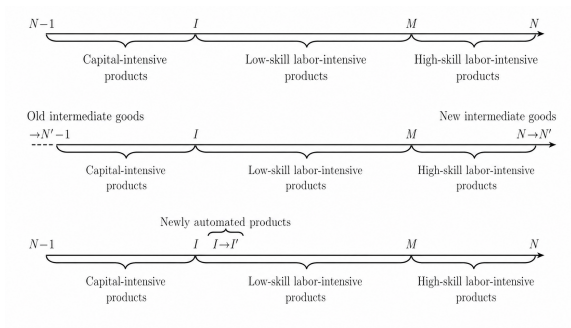
Task Measures and State Variables

AI advancement measure:

$$n(t) = N(t) - I(t)$$

Low-skill task measure:

$$m(t) = M(t) - I(t)$$



Assumption 3: Functional Forms

$$\gamma_H(i, t) = e^{A_H i}, \quad \gamma_L(i, t) = \Gamma(t) e^{A_L i}$$

$$A_H > A_L$$

Properties:

- ▶ Exponential productivity growth across tasks
- ▶ High-skill advantage increases with i

Role of $\Gamma(t)$:

- ▶ Captures endogenous learning of low-skill workers

Learning Mechanism

$$\Gamma(t) = \beta E(t) l_T(t)^\alpha l_P(t)^{1-\alpha}$$

$$l_T(t) + l_P(t) = l_L(t)$$

$$l_T(t) = \alpha l_L(t), \quad l_P(t) = (1 - \alpha) l_L(t)$$

Interpretation:

- ▶ $E(t)$: exposure to AI technologies
- ▶ l_T : training time
- ▶ l_P : production experience

Implication:

- ▶ Productivity gap narrows over time
- ▶ Learning acts as an equalizing force

Household Problem: Low-Skill Workers

Objective:

$$\max \int_0^{\infty} e^{-\rho t} [\ln c_L(t) + \ln(1 - l_L(t))] dt$$

Budget constraint:

$$\dot{a}_L(t) = (R(t) - g(t))a_L(t) + \theta w_L(t)l_T(t) + w_L(t)l_P(t) - c_L(t)$$

- ▶ Training generates partial income
- ▶ θ : training cost parameter

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Budget constraint:

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- ▶ High-skill workers benefit from training sector
- ▶ Captures redistribution via knowledge transfer

Equilibrium Intuition

- ▶ Automation $\uparrow \Rightarrow$ inequality $\uparrow$
- ▶ Task creation $\uparrow \Rightarrow$ demand for skills $\uparrow$
- ▶ Learning $\uparrow \Rightarrow$ inequality $\downarrow$

Next Steps

- ▶ Solve Balanced Growth Path
- ▶ Endogenize AI technology growth
- ▶ Quantitative analysis